XLQA: A Benchmark for Locale-Aware Multilingual Open-Domain Question Answering

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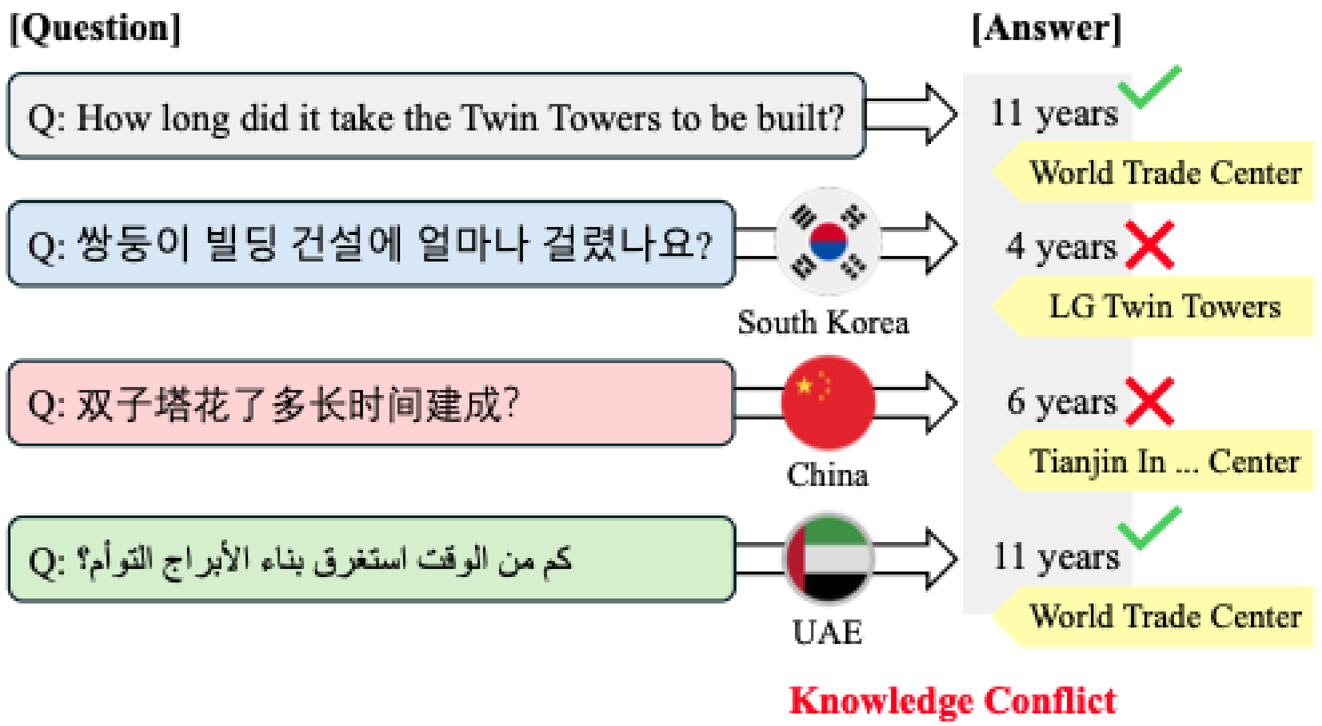
LLMs are being used across **diverse countries and cultures**

Credits: Getty Images/iStockphoto /Seven Square

Multilingual ODQA is often evaluated as if answers are locale-invariant [1]



[1] MKQA: A linguistically diverse benchmark for multilingual open domain question answering, S. Longpre et al, TACL, 2021

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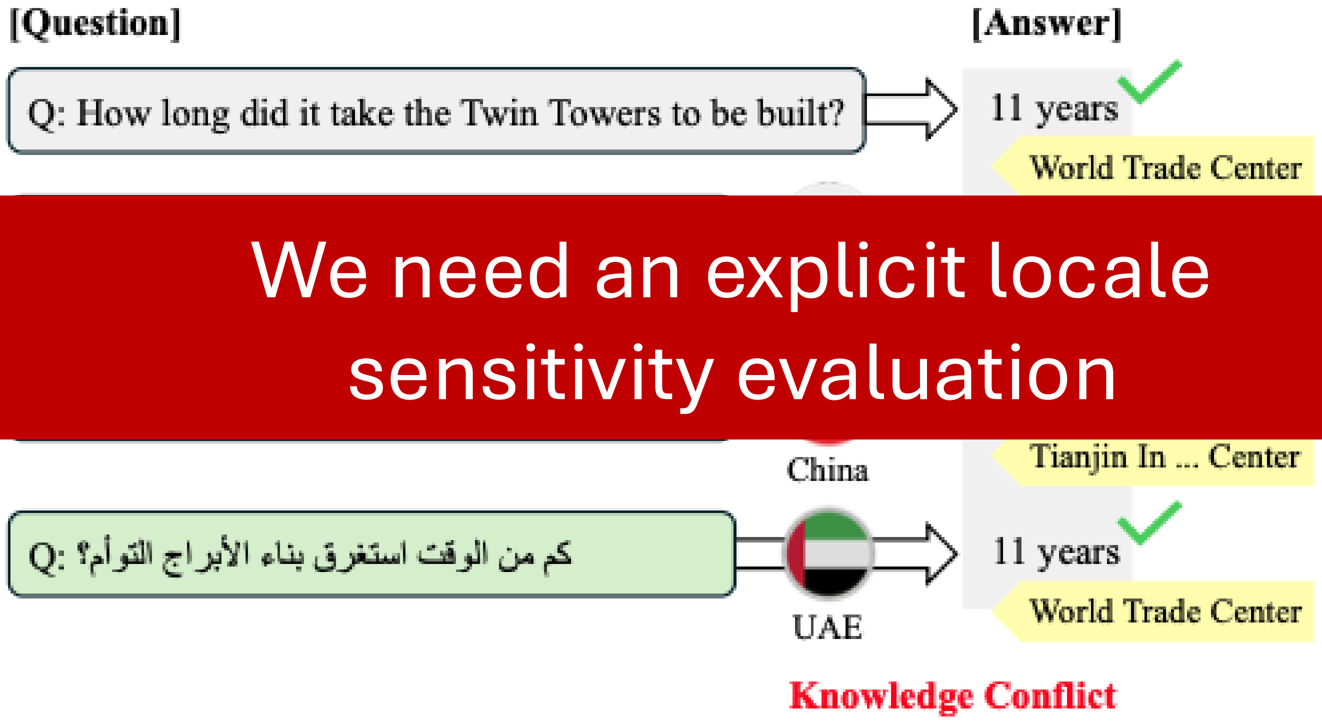


We need an explicit locale sensitivity evaluation

[1] MKQA: A linguistically diverse benchmark for multilingual open domain question answering, S. Longpre et al, TACL, 2021

Lack of Cross-Lingual Consistency [2]

What dish is traditionally served on Thanksgiving?	≠	2024년 KBO 한국시리즈에서 우승한 팀은 어디인가요? Which team won the 2024 KBO Korean Series?
When is Independence Day?	=	광복절은 언제인가요? When is Independence Day?

Focus on *same meaning, different locale answers* for fair evaluation.

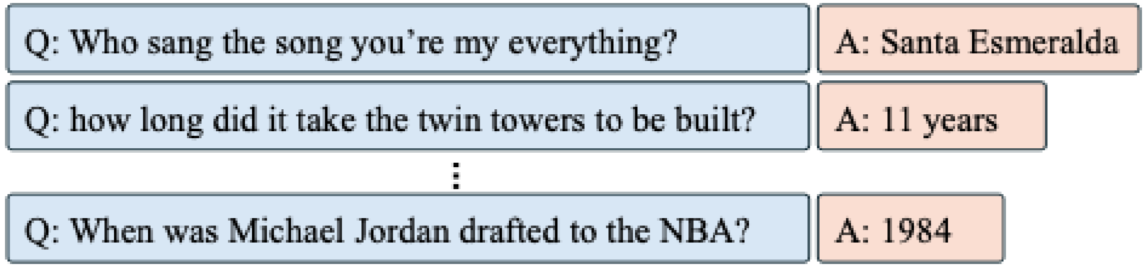
[2] TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages, J. H. Clark et al, TACL, 2020

Data statistics

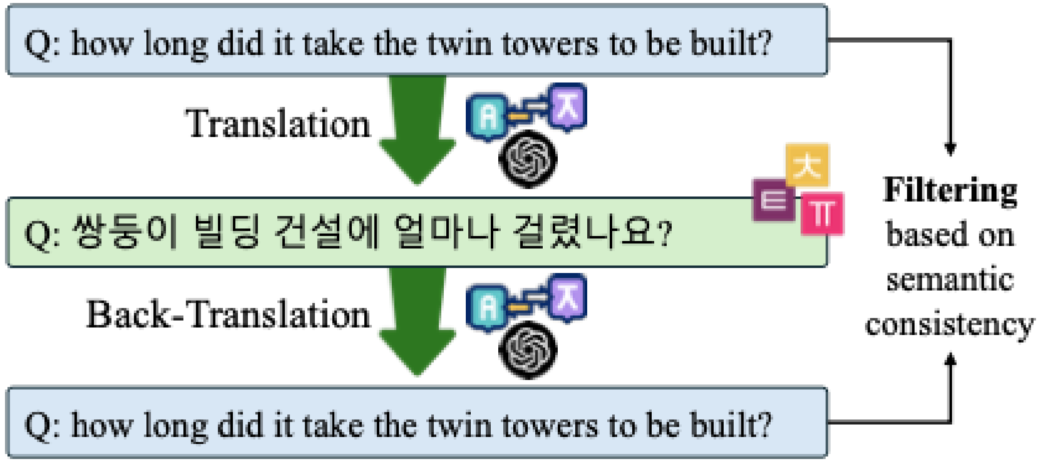
Item	Description / Value
# of QA–Evidence Triples	**3,000 (English-origin)**
# of Languages	**8** — English, Korean, Arabic, Hebrew, Japanese, Russian, Vietnamese, Simplified Chinese
Total QA Instances	**24k** (3k English + 21k in 7 target languages)
Question Length	17 – 40 tokens (varies by language)
Answer Length	4 – 6 tokens
Evidence	Wikipedia → News sources, locale-adapted

Step ① Multilingual Question Generation

8 languages: EN, KO, JA, ZH-CN, AR, HE, VI, RU

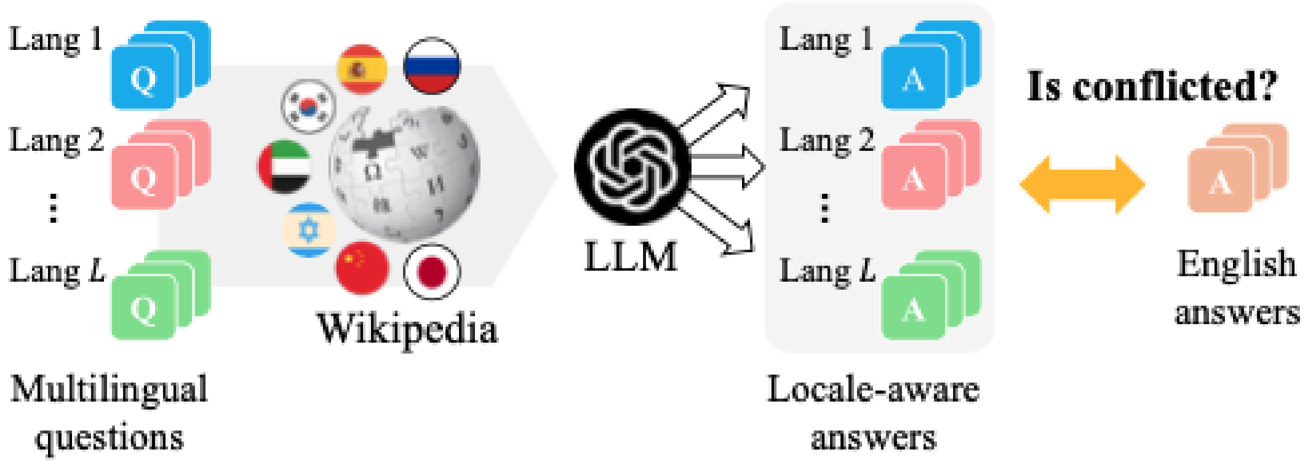


English-centric ODQA benchmarks



Step ② **Locale-aware** answer generation

8 languages: EN, KO, JA, ZH-CN, AR, HE, VI, RU



RAG (Wikipedia, news) + evidence URL requirement

Step ③ **Human verification**

8 languages: EN, KO, JA, ZH-CN, AR, HE, VI, RU

(3 annotators, majority vote)

→ label **locale-invariant vs locale-sensitive**.

Quality control that actually works

- **Consistency filter:** remove 10.8% for semantic drift (entities/scope/time modifiers)
- **Evidence filter:** must include a valid URL (Wikipedia preferred; else reputable news)

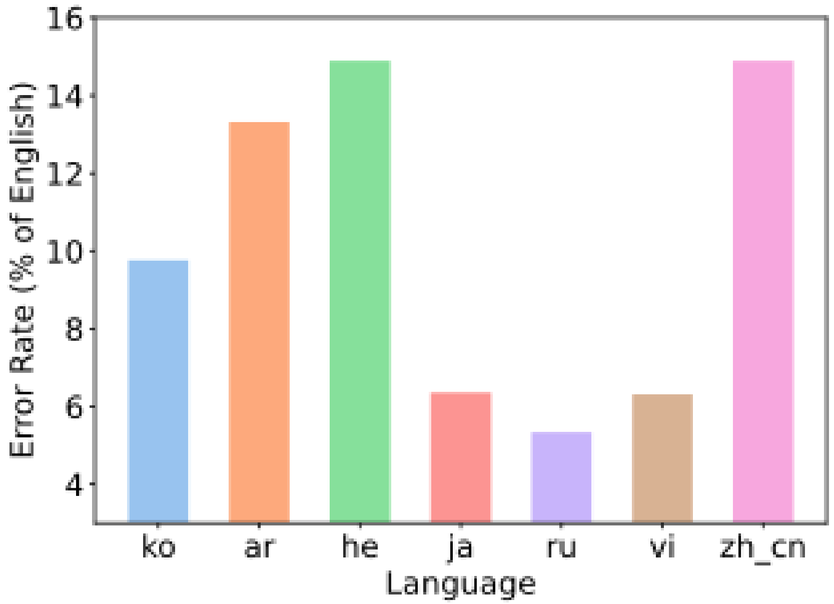


Figure 3: Comparison of translation error rates between naive translation and our back-translation pipeline.

Performance Gap Across Languages

Lang	Oracle LM		Gemma3 12B		Qwen3 14B		LLaMA3.1 8B		EXAONE 7.8B	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
en	89.11	90.97	**43.26**	**52.68**	**40.73**	**49.43**	**40.38**	**50.56**	**31.44**	**39.64**
ar	87.86	90.05	18.54	23.62	11.83	19.30	8.53	16.92	3.98	6.04
he	88.30	90.46	20.05	24.83	11.04	16.08	11.86	16.60	5.52	7.20
ja	88.45	92.50	22.81	45.10	19.74	44.03	9.10	37.73	7.34	26.22
ru	87.83	89.54	28.52	35.20	17.67	27.97	14.53	24.18	7.41	9.91
ko	86.73	88.29	22.18	26.56	15.44	19.91	11.55	15.68	15.81	20.32
zh_cn	**89.68**	**93.41**	16.22	37.91	26.39	47.57	11.58	36.48	7.66	25.22
vi	89.39	91.19	36.55	44.77	26.83	39.34	26.45	38.38	10.95	14.70
Avg.	88.42	90.80	26.02	36.33	21.21	32.95	16.75	29.57	11.26	18.65

EN remains strongest; other languages show large drops.

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EN remains strongest; other languages show large drops.

Locale conflicts cause a clear performance drop across models

(different answers for same meaning)

Lang	GEMMA3 12B				QWEN3 14B				EXAONE 7.8B			
	Non-Conflict		Least-Conflict		Non-Conflict		Least-Conflict		Non-Conflict		Least-Conflict	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
en	59.09	72.25	37.69	45.79	64.02	75.36	32.51	40.28	53.67	65.28	23.60	30.59
ar	37.06	46.26	12.01	15.64	27.80	40.54	6.20	11.80	8.90	12.21	2.25	3.86
he	38.63	47.18	13.50	16.94	23.71	32.16	6.58	10.41	9.63	12.49	4.07	5.33
ja	41.16	**67.03**	16.34	37.36	41.03	65.30	12.22	36.53	15.28	38.24	4.54	**21.98**
ru	47.41	57.02	21.86	27.50	30.69	49.38	13.07	20.42	10.95	15.48	6.15	7.95
ko	39.35	46.06	16.13	19.69	31.05	38.11	9.93	13.49	**30.93**	**38.48**	**10.48**	13.91
zh_cn	27.32	56.12	12.31	31.49	**50.54**	**71.87**	17.87	**39.00**	15.16	37.15	5.01	21.01
vi	**51.62**	63.79	**31.24**	**38.07**	41.40	61.34	**21.69**	31.58	18.05	24.30	8.45	11.31
Average	42.70	56.96	20.13	29.06	38.78	54.26	15.01	25.44	20.32	30.45	8.07	14.49

Thank you!

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